CSCI 1101 Introduction to Computer Science

Data Representation

Overview

Definitions

- Data
- Data Representation

Number Systems

- Binary
- Decimal
- Hexadecimal

Conversions

- From Decimal
- From Binary
- From Hexadecimal

Computers & Data

- Representing Numbers
- Representing Text
- Representing Images
- Representing Sound
- Representing Video

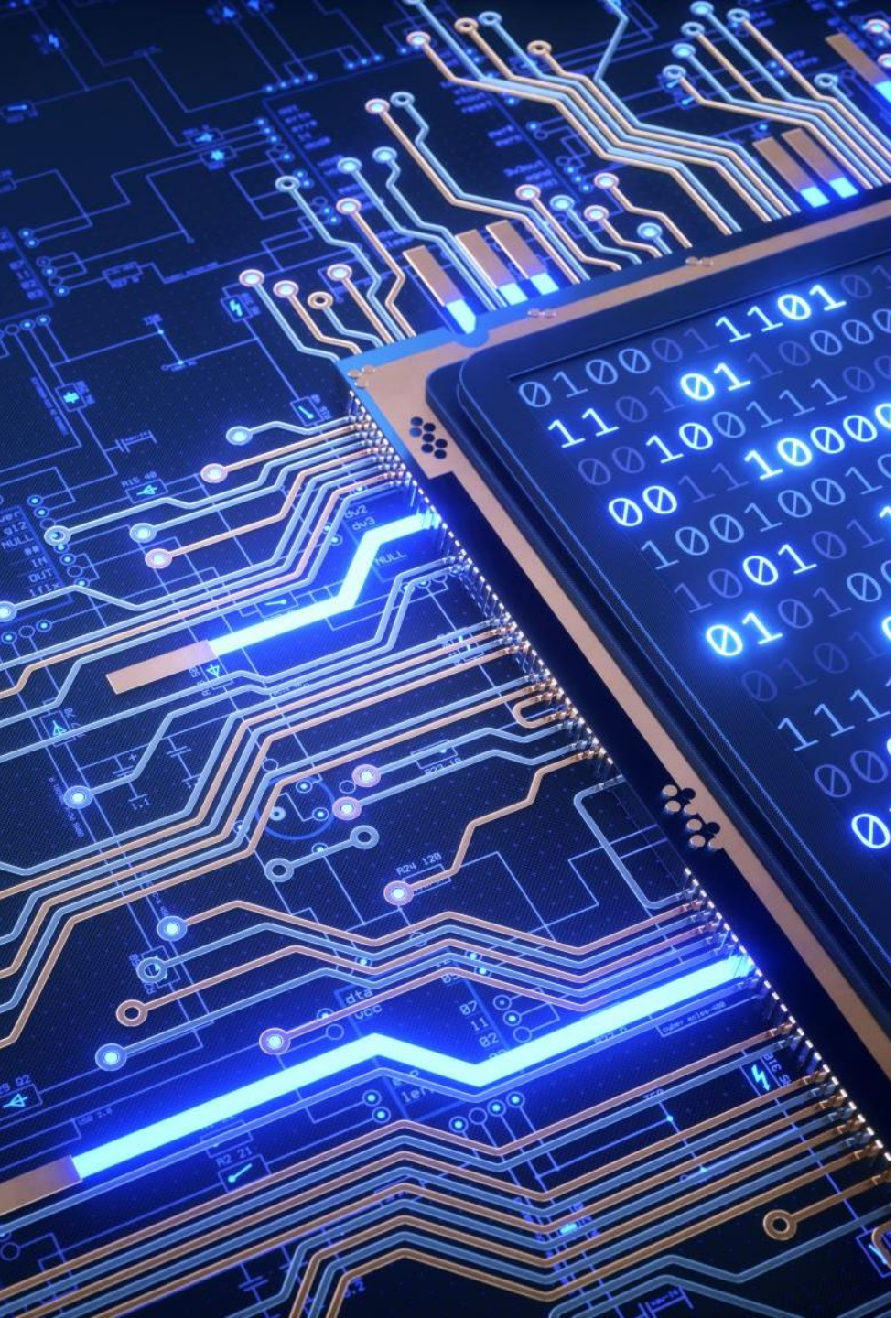
Definitions

What is Data?

- **Data** refers to any collection of raw facts, figures, or information that can be processed, stored, and analyzed.
- Thus, data can be a:
  - Numbers
  - text
  - images
  - Audio
  - Video
  - other types of input

Computers and Data

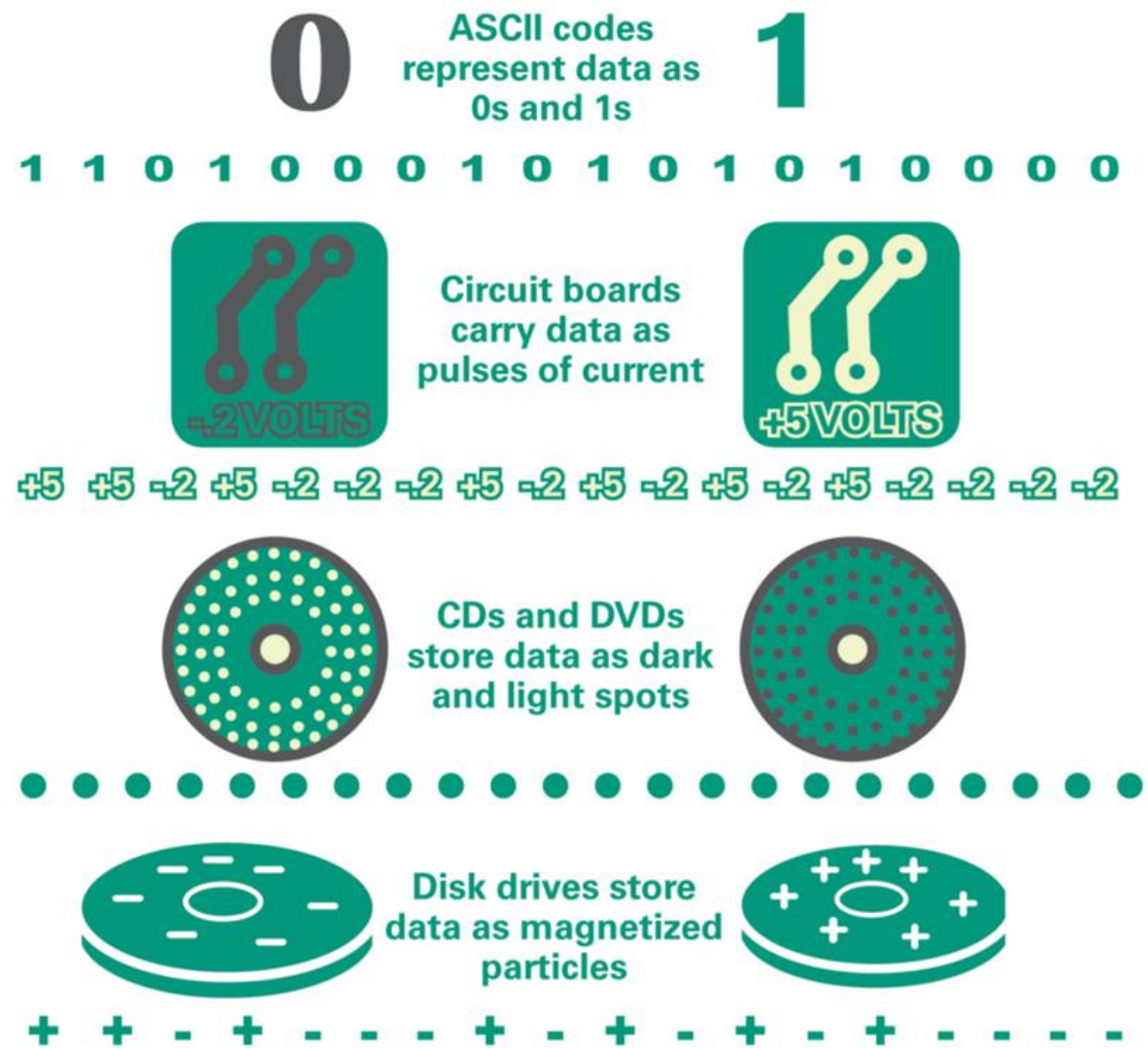
- Data Representation refers to the form in which data is stored, processed, and transmitted.
- Music, videos, games, and other information are the computer's data.
- Data is stored in digital (data expressed in binary) formats that can be handled by electronic circuitry.



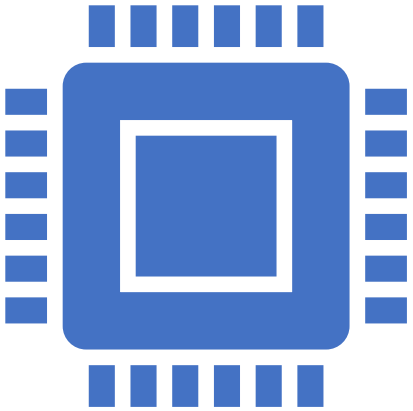
Why Binary?

- Computers operate with voltage through what amounts to microscopic switches (transistors) that are either “on” (1) or “off” (0).

Low Voltage = 0
Hight Voltage = 1

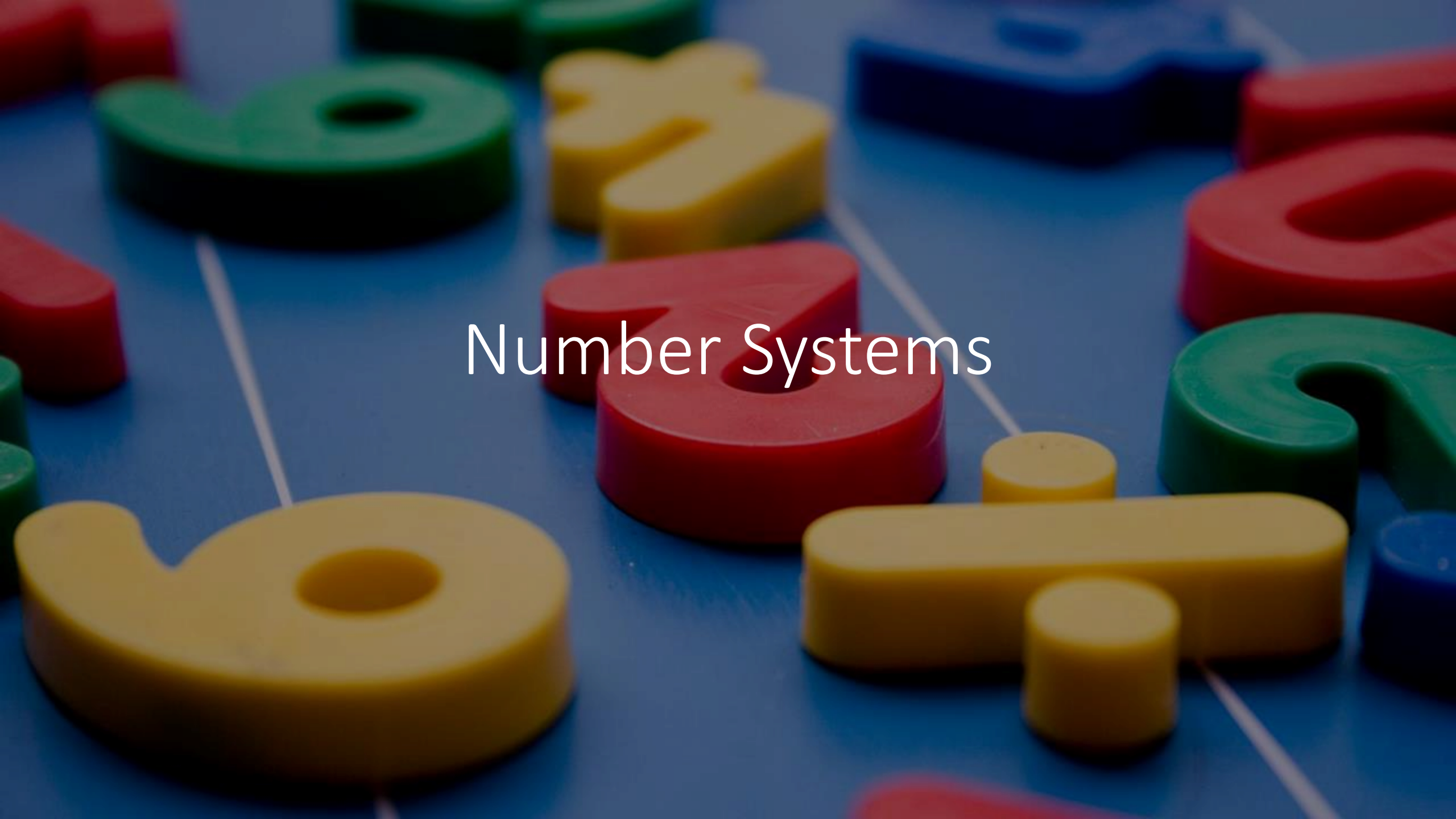

0
ASCII codes represent data as 0s and 1s
1
1 1 0 1 0 0 0 1 0 1 0 1 0 1 0 0 0 0
-2VOLTS
Circuit boards carry data as pulses of current
+5VOLTS
+5 +5 -2 +5 -2 -2 -2 +5 -2 +5 -2 +5 -2 +5 -2 -2 -2 -2
CDs and DVDs store data as dark and light spots
Disk drives store data as magnetized particles
+ + - + - - - + - + - + - + - - - -

Bits and Bytes



- Terminology related to bits and bytes is extensively used to describe storage capacity and network access speed.
  - Bits → binary digit
  - Byte → group of eight bits (abbreviated as an uppercase B)
- Have you ever heard phrases like “megabytes”, “terabytes”, or something similar?

Unit	Abbreviation	Approximate Size
Bit	b	Binary digit, single 1 or 0
Nibble	–	4 bits
Byte	B	8 bits
Kilobyte	KB	1,024 bytes or 10^3
Megabyte	MB	1,024 KB or 10^6
Gigabyte	GB	1,024 MB or 10^9
Terabyte	TB	1,024 GB or 10^{12}
Petabyte	PB	1,024 TB or 10^{15}
Exabyte	EB	1,024 PB or 10^{18}
Zettabyte	ZB	1,024 EB or or 10^{21}



Number Systems

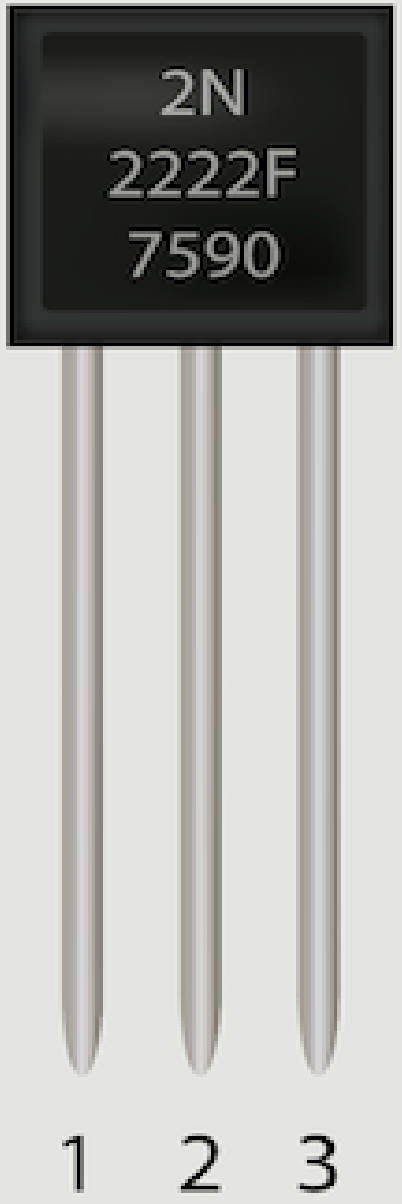
Decimal Numbers

- Think about the numbers, you learned at school while growing up.
- They had 10 digits: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9
- The number 1,248 is
$$1{,}000 + 200 + 40 + 8$$
- We can change the representation to:
$$(1 * 1{,}000) + (2 * 100) + (4 * 10) + (8 * 1)$$
- We can change it even further to:
$$(1 * 10^3) + (2 * 10^2) + (4 * 10^1) + (8 * 10^0)$$

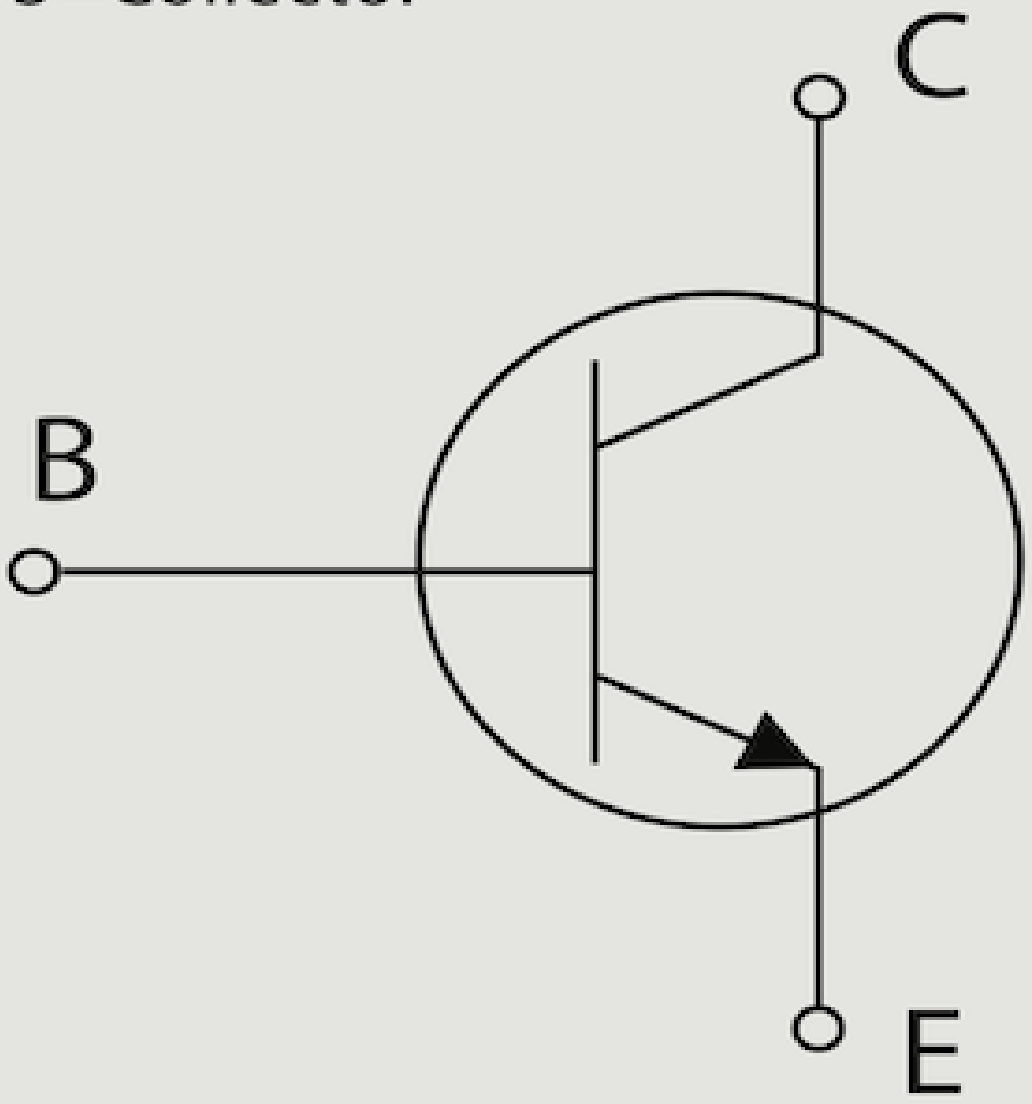
Deca

- Digits: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9
- Base 10

10^3	10^2	10^1	10^0



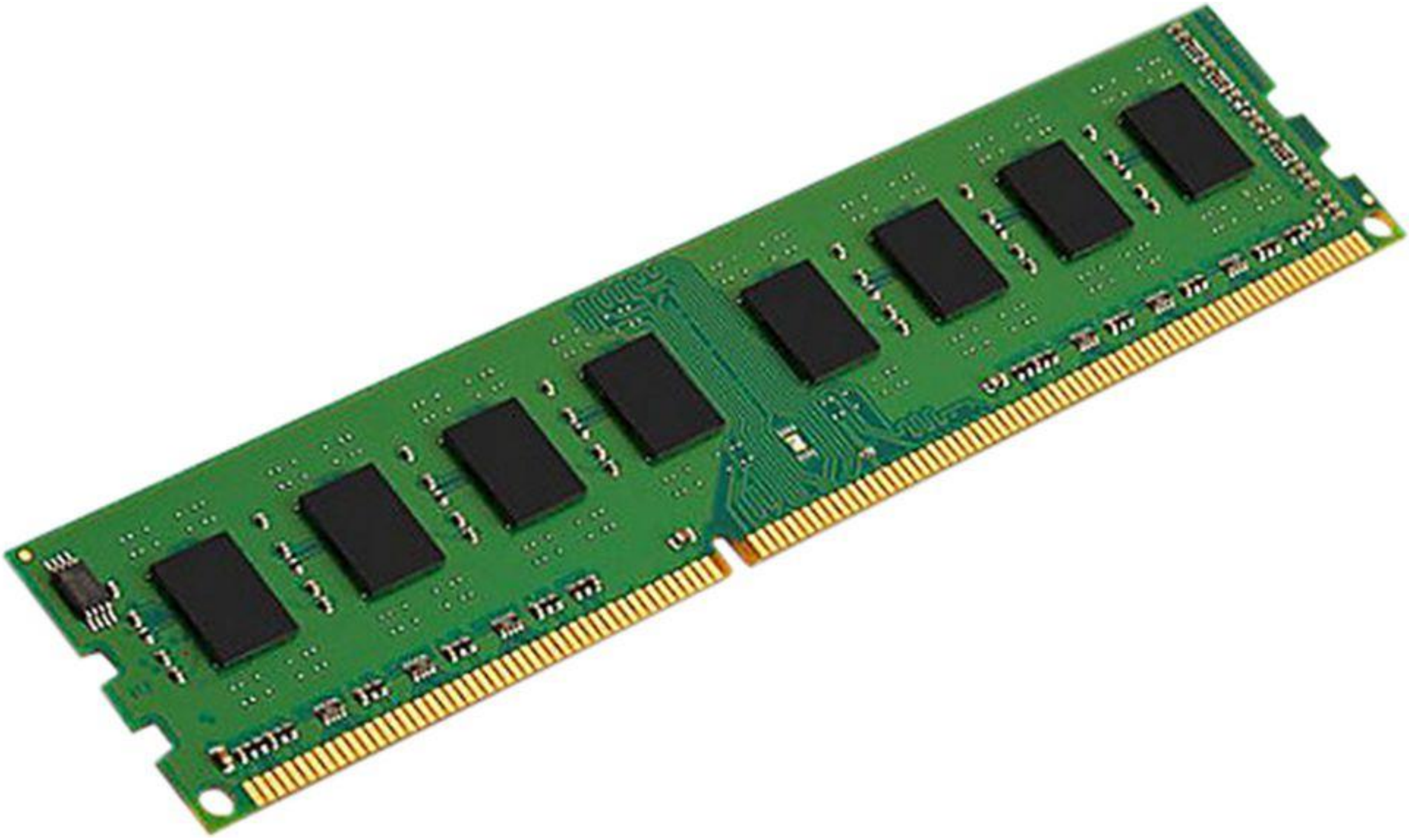
1=Emitter
2=Base
3=Collector



Binary

- Digits: 0, 1
- Base 2
- Binary digit → *bit*

2^3	2^2	2^1	2^0



Hexadecimal

- Human-friendly way of grouping binary data
- Digits: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F
- Base 16
- Every hexadecimal digit corresponds to four binary bits.

```
1 = 0001
A = 1010
F = 1111
```

16^3	16^2	16^1	16^0

Summary

Decimal (Base 10)	Binary (Base 2)	Hexadecimal (Base 16)
0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7
8	1000	8
9	1001	9
10	1010	A
11	1011	B
12	1100	C
13	1101	D
14	1110	E
15	1111	F

Addition and Subtraction

Addition

	0	0	0	1
+	0	0	0	0
	0	0	0	1

			1	
	0	0	0	1
+	0	0	0	1
	0	0	1	0

Addition (Example)

$0110_2 + 0011_2$

	1	1		
	0	1	1	0
+	0	0	1	1
	1	0	0	1

Addition (Your turn)

00101_2 + 01111_2

0	0	1	0	1
+ 0	1	1	1	1
1	0	1	0	0

Subtraction

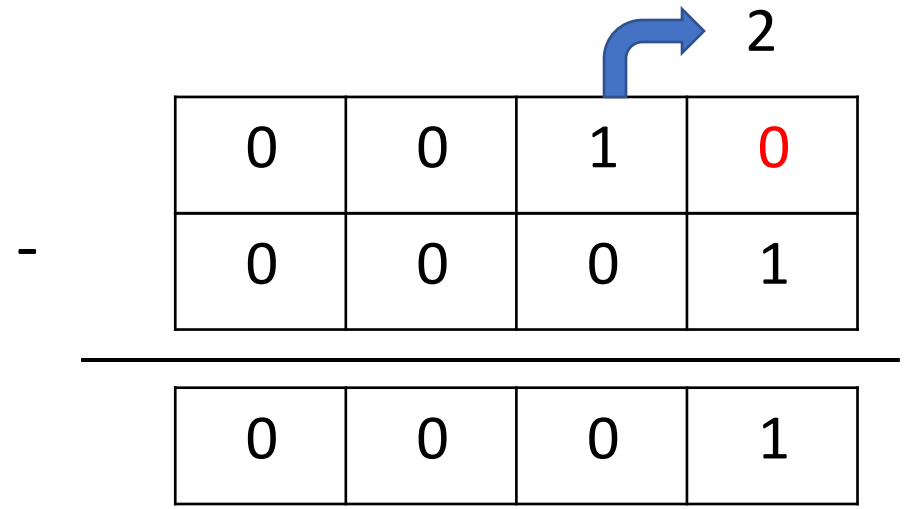
	0	0	0	1
-	0	0	0	0
	0	0	0	1

	0	0	0	1
-	0	0	0	1
	0	0	0	0

Subtraction

RULES:

1. We borrow from the column to the left.
2. If 0, we continue to the next column
3. If 1, this number becomes 0 and 2 is passed to the column to the right



Subtraction (Example)

0110_2 - 0011_2

	0	1	1	0
-	0	0	1	1
	0	0	1	1

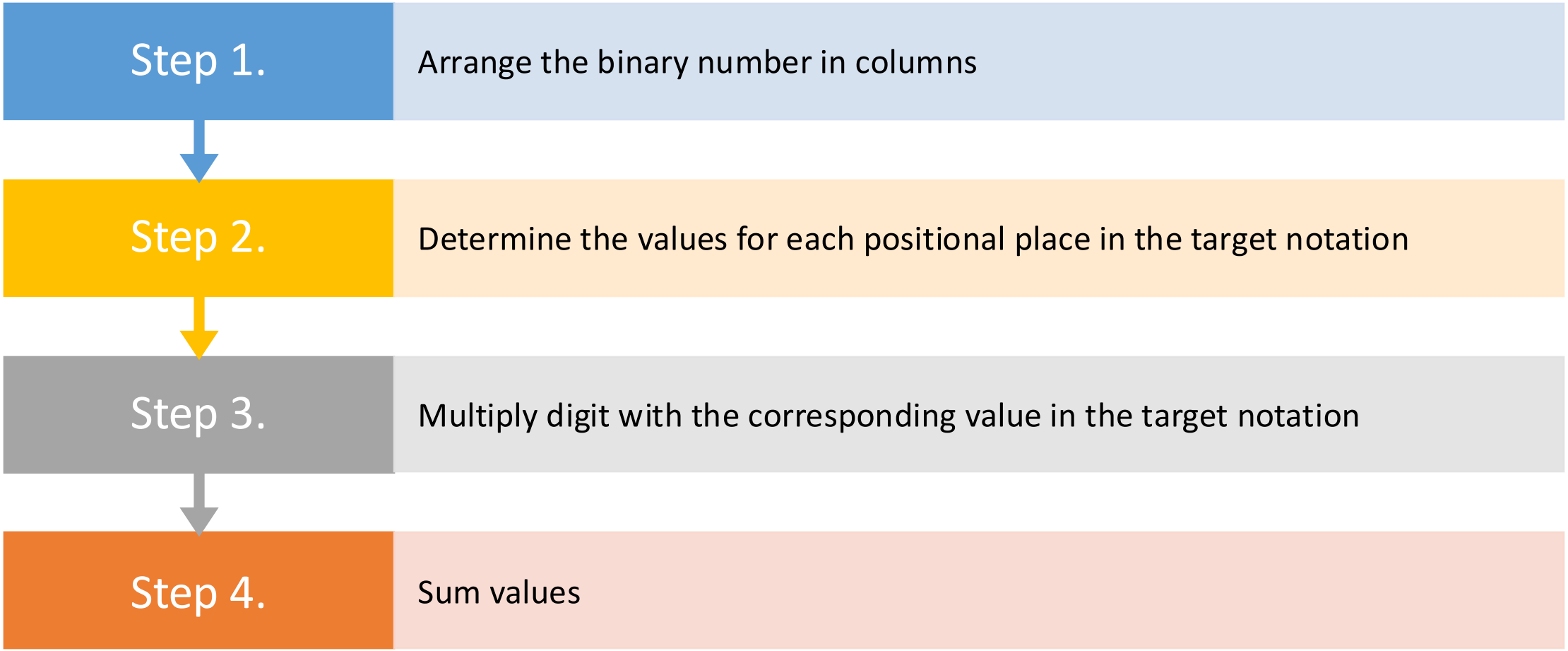
Subtraction (Example)

1100_2 - 0111_2

	1	1	0	0
-	0	1	1	1
	0	1	0	1

Conversions

Conversions to Decimal (Method 1)



Start here →

Example
Binary to Dec

$10110011_2 = X_{10}$

1	0	1	1	0	0	1	1
128 2^7	64 2^6	32 2^5	16 2^4	8 2^3	4 2^2	2 2^1	1 2^0
128		32	16			2	1

sum →

$10110011_2 = 179_{10}$

Start here →

Example
Hex to Dec

$A2F7_{16}$ = X_{10}

A	2	F	7
10	2	15	7
4096_{16^3}	256_{16^2}	16_{16^1}	1_{16^0}
40960	512	240	7

sum →

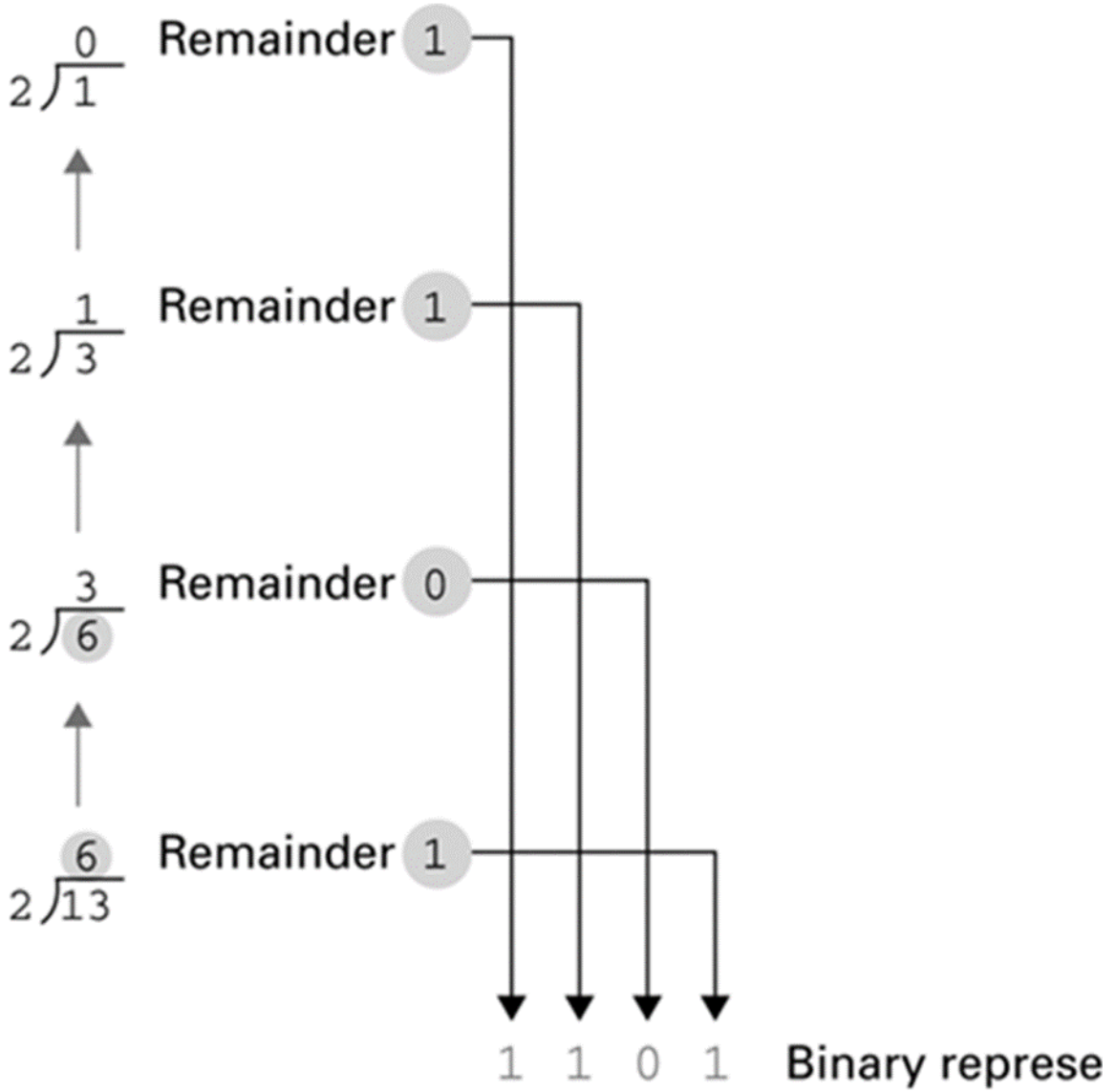
$A2F7_2$ = 41719_{10}

Converting to Decimal (Method 2)

Step 1.	Divide the value by the desired base number (B) and record the remainder.
Step 2.	As long as the quotient obtained is not zero, continue to divide the newest quotient by B and record the remainder.
Step 3.	Now that a quotient of zero has been obtained, the new number system representation of the original value consists of the remainders listed from right to left in the order they were recorded.

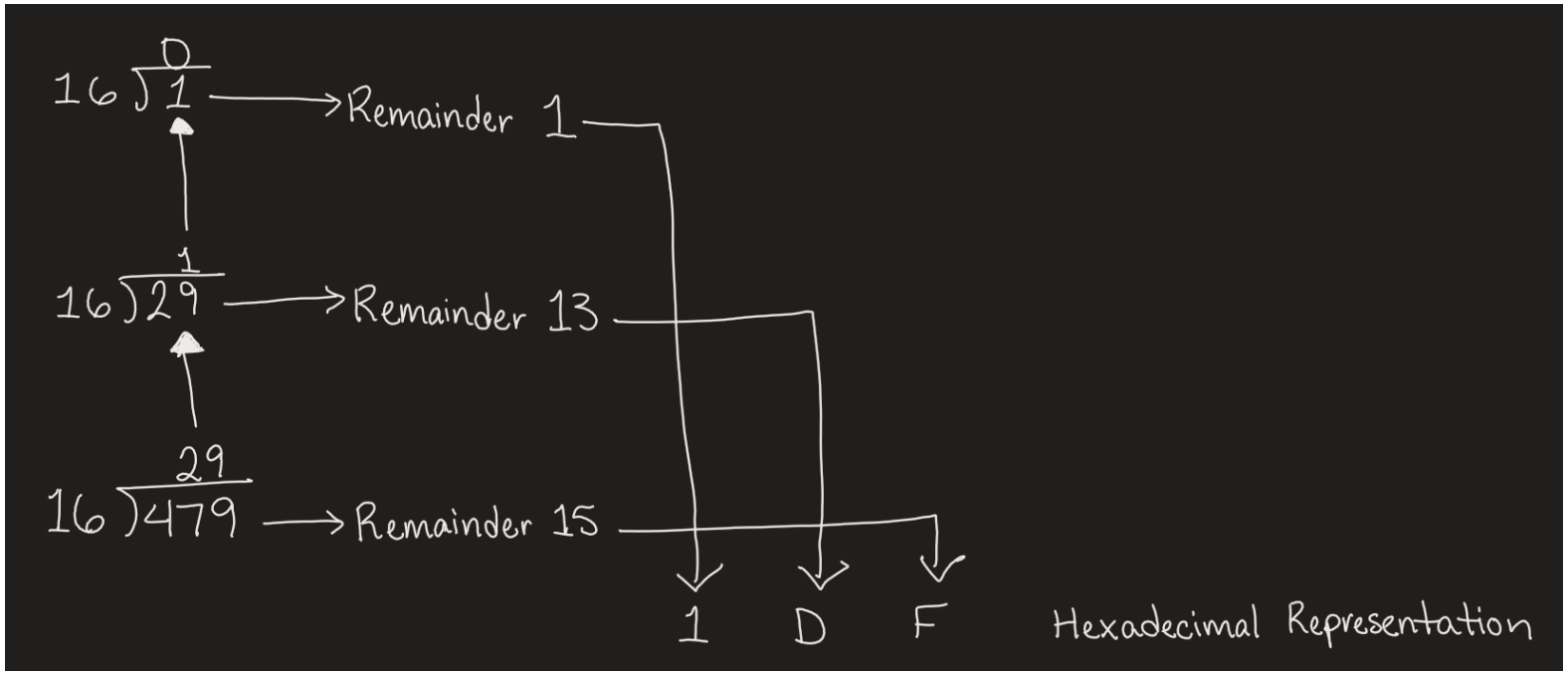
Example 1-1

Start here →



Example 1-2

Start here →

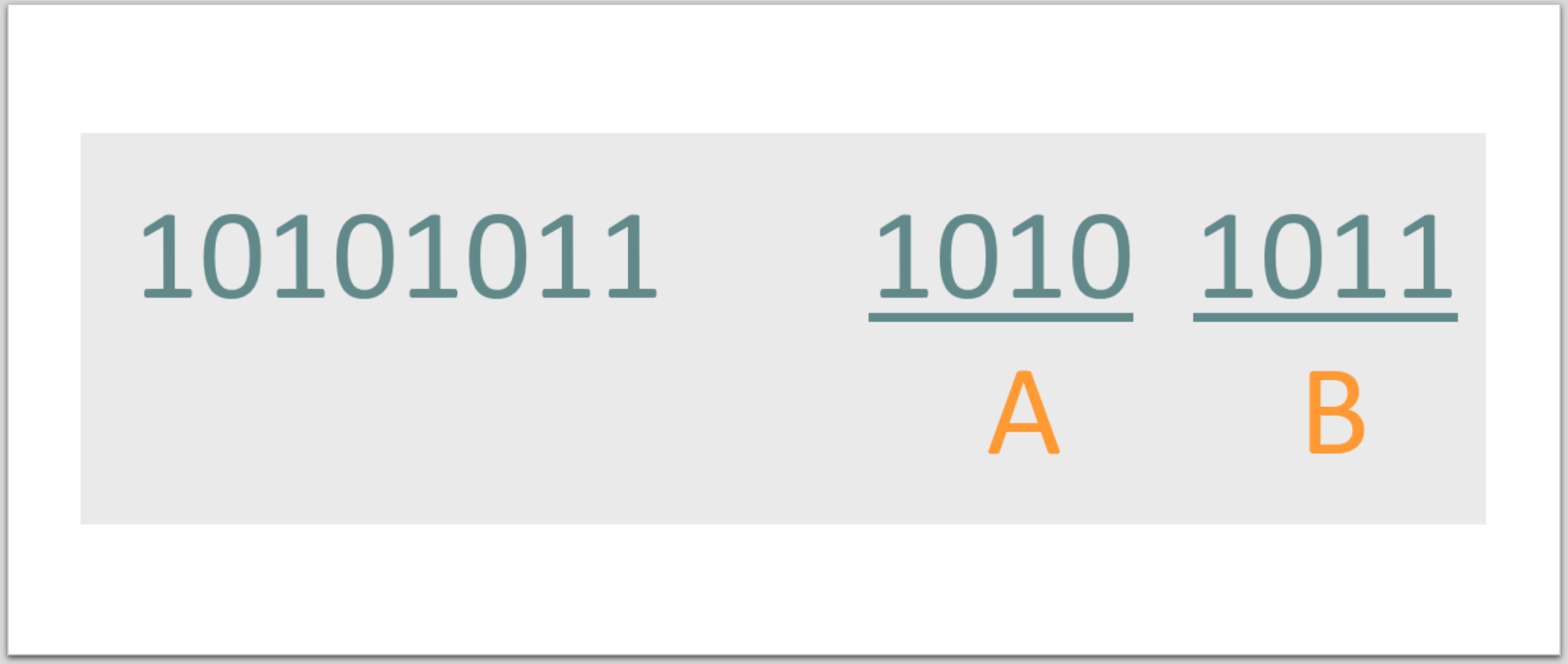


Converting From Binary To Hexadecimal

Step 1. Mark groups of four bits (starting at the right end)

Step 2. Convert each group of bits into hex digits.

Bit pattern	Hexadecimal representation
0000	0
0001	1
0010	2
0011	3
0100	4
0101	5
0110	6
0111	7
1000	8
1001	9
1010	A
1011	B
1100	C
1101	D
1110	E
1111	F



Example 3

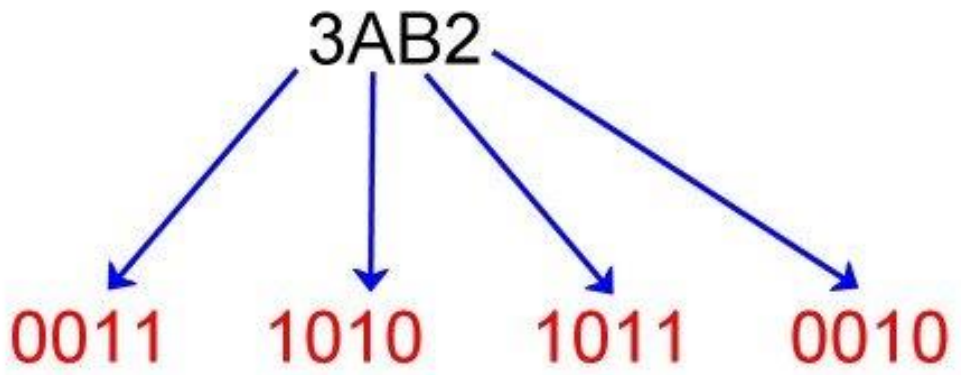
Converting From Hexadecimal To Binary

Step 1. Convert each hex digit into binary from left to right

Bit pattern	Hexadecimal representation
0000	0
0001	1
0010	2
0011	3
0100	4
0101	5
0110	6
0111	7
1000	8
1001	9
1010	A
1011	B
1100	C
1101	D
1110	E
1111	F

Example 4

Converting Hex to Binary



$3AB2_{16} = 11101010110010_2$

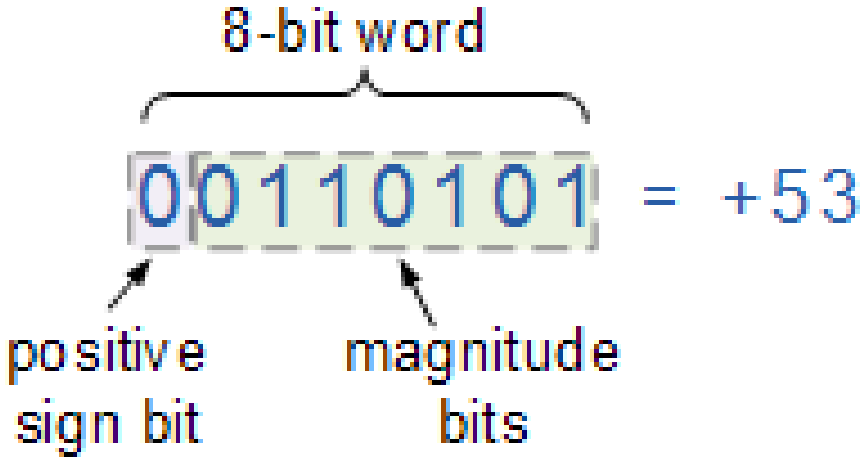
Representing Numbers

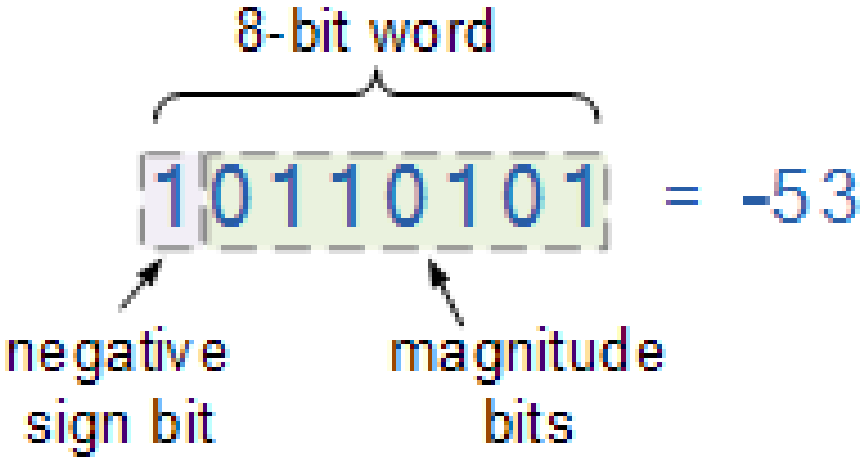
Representing Numeric Values

- Using the number system (binary, hexadecimal), whole numbers (0, 1, 2,…) can be represented
- What about integers and floating-point (real) numbers?
  - Modify binary system to represent these values as well

Representing Negative Values

- Signed-Magnitude (SM) Notation
- 1’s Complement
- 2’s Complement





Sign-Magnitude (SM) Notation

The leftmost bit (**most significant bit**) represents the sign:

- positive (0) or
- negative (1).

The rest of the binary digits are used to represent the magnitude of the binary number.

What is the problem with SM Notation?

There are two representations for 0, a positive and negative representation.

1’s Complement

Positive numbers remain unchanged and start with a 0.
Negative numbers are obtained by inverting the unsigned positive number (changing the 0 to 1 and 1 to 0).

Example: 8-bit representation of $\pm$12

+12: `0000 1100`

-12: `1111 0011`

What is the problem with 1’s complement?

It also has two representations for 0.

+0: `0000`

- 0: `1111`

2’s Complement

Two's Complement is the most widely used method for representing signed integers (both positive and negative) in binary.

Two's Complement = One's Complement + 1

Example: 8-bit representation of ±12

```
+12:  0000 1100
-12:  1111 0011   // one’s complement
  1:  0000 0001   // plus one
-12:  1111 0100
```

There is a single representation for 0 with 2’s Complement.

Representing Real Numbers

- Consider weights and placement
  - Decimal System
    - Examples: 104.32, 0.99999, 357.0, and 3.14159
    - Weights to the right of the decimal are: 10^{-1}, 10^{-2}, 10^{-3}, …
  - Binary System
    - Examples: 10.101, 0.11101, 1011.1, and 11.111
    - Weights to the right of the decimal are: 2^{-1}, 2^{-2}, 2^{-3}, …

Converting To Decimal

Step 1. Multiply the digit by the quantity it represents (B^n) and record the value. It is recommended to start from right end.

Step 2. Repeat step 1 until no digits in the number given are left.

Step 3. Add all the recorded values together to obtain the equivalent decimal value.

Converting Binary Fractions to Decimal Fractions

- Same process – multiply each fractional bit by 2 (base) raised to the corresponding power and add the products together
- Example:

$(0.101)_2 = (\quad)_{10}$

1	0	1	0	0
$^1/_2$ 2^{-1}	$^1/_4$ 2^{-2}	$^1/_8$ 2^{-3}	$^1/_{16}$ 2^{-4}	$^1/_{32}$ 2^{-5}

$$1 * 2^{-1} + 0 * 2^{-2} + 1 * 2^{-3}$$

This is equivalent to

$$1 * {}^1/_2 + 0 * {}^1/_4 + 1 * {}^1/_8 = 0.625$$

Thus, $(0.101)_2 = (0.625)_{10}$

Real Numbers in Programming

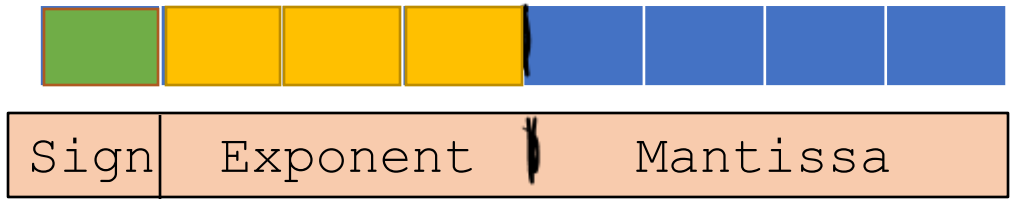
Value	Stored as
Float	sign bit, 8-bit exponent, 23-bit mantissa
Double	sign bit, 11-bit exponent, 52-bit mantissa

Storing Real Numbers

- Storing real numbers in binary is done using a method called **floating-point representation**
- This breaks a number into three parts:
  - the **sign**,
  - the **mantissa (or significand)**,
  - and the **exponent**.
- This format allows computers to store very large or very small real numbers efficiently.

Storing Real Numbers

- Sign: A single bit: 0 for positive, 1 for negative
- Mantissa: In binary, the mantissa is normalized to always start with 1. followed by the fractional part. Only the fractional part is stored (since the leading 1. is implicit in the normalized form).
- Exponent: The exponent determines how many places to move the binary point, effectively scaling the mantissa. It is stored using biased representation.



IEEE 754 Standard (single precision)

In IEEE 754 (the standard used in most systems for floating-point arithmetic), a 32-bit number is divided into:

1 bit for the sign

8 bits for the exponent, with a bias of 127

23 bits for the mantissa, storing the fractional part after normalization.

Example: $(10.75)_{10} = (1010.11)_2$

Normalized number = 1010.11 becomes 1.01011×2^3

Mantissa = 01011000000000000000000 (23 bits, fractional part only).

Exponent = Since the exponent is 3, we store 3 + 127 = 130 in binary as 10000010.

IEEE 754 Standard Representation: 01000001001011000000000000000000

1 Sign + 8 Exponent + 23 Mantissa = 32 bits

Representing Text

https://youtu.be/TpoV_iVhFt0?si=YvdN0uZ67kNIq5YC

ASCII	Unicode	Extended ASCII
7 bits per character	16, 24, or 32 bits per character	8 bits per character
128 different characters	Between 65,536 – 4,294,967,296 characters depending on the system ASCII	256 different characters
All English letters, numbers, and symbols	Contains all major languages, symbols, hieroglyphics, etc.	

ASCII Code

Decimal	Hex	Char	Decimal	Hex	Char	Decimal	Hex	Char	Decimal	Hex	Char
0	0	[NULL]	32	20	[SPACE]	64	40	@	96	60	`
1	1	[START OF HEADING]	33	21	!	65	41	A	97	61	a
2	2	[START OF TEXT]	34	22	"	66	42	B	98	62	b
3	3	[END OF TEXT]	35	23	#	67	43	C	99	63	c
4	4	[END OF TRANSMISSION]	36	24	$	68	44	D	100	64	d
5	5	[ENQUIRY]	37	25	%	69	45	E	101	65	e
6	6	[ACKNOWLEDGE]	38	26	&	70	46	F	102	66	f
7	7	[BELL]	39	27	'	71	47	G	103	67	g
8	8	[BACKSPACE]	40	28	(	72	48	H	104	68	h
9	9	[HORIZONTAL TAB]	41	29	)	73	49	I	105	69	i
10	A	[LINE FEED]	42	2A	*	74	4A	J	106	6A	j
11	B	[VERTICAL TAB]	43	2B	+	75	4B	K	107	6B	k
12	C	[FORM FEED]	44	2C	,	76	4C	L	108	6C	l
13	D	[CARRIAGE RETURN]	45	2D	-	77	4D	M	109	6D	m
14	E	[SHIFT OUT]	46	2E	.	78	4E	N	110	6E	n
15	F	[SHIFT IN]	47	2F	/	79	4F	O	111	6F	o
16	10	[DATA LINK ESCAPE]	48	30	0	80	50	P	112	70	p
17	11	[DEVICE CONTROL 1]	49	31	1	81	51	Q	113	71	q
18	12	[DEVICE CONTROL 2]	50	32	2	82	52	R	114	72	r
19	13	[DEVICE CONTROL 3]	51	33	3	83	53	S	115	73	s
20	14	[DEVICE CONTROL 4]	52	34	4	84	54	T	116	74	t
21	15	[NEGATIVE ACKNOWLEDGE]	53	35	5	85	55	U	117	75	u
22	16	[SYNCHRONOUS IDLE]	54	36	6	86	56	V	118	76	v
23	17	[ENG OF TRANS. BLOCK]	55	37	7	87	57	W	119	77	w
24	18	[CANCEL]	56	38	8	88	58	X	120	78	x
25	19	[END OF MEDIUM]	57	39	9	89	59	Y	121	79	y
26	1A	[SUBSTITUTE]	58	3A	:	90	5A	Z	122	7A	z
27	1B	[ESCAPE]	59	3B	;	91	5B	[	123	7B	{
28	1C	[FILE SEPARATOR]	60	3C	<	92	5C	\	124	7C	|
29	1D	[GROUP SEPARATOR]	61	3D	=	93	5D	]	125	7D	}
30	1E	[RECORD SEPARATOR]	62	3E	>	94	5E	^	126	7E	~
31	1F	[UNIT SEPARATOR]	63	3F	?	95	5F	_	127	7F	[DEL]

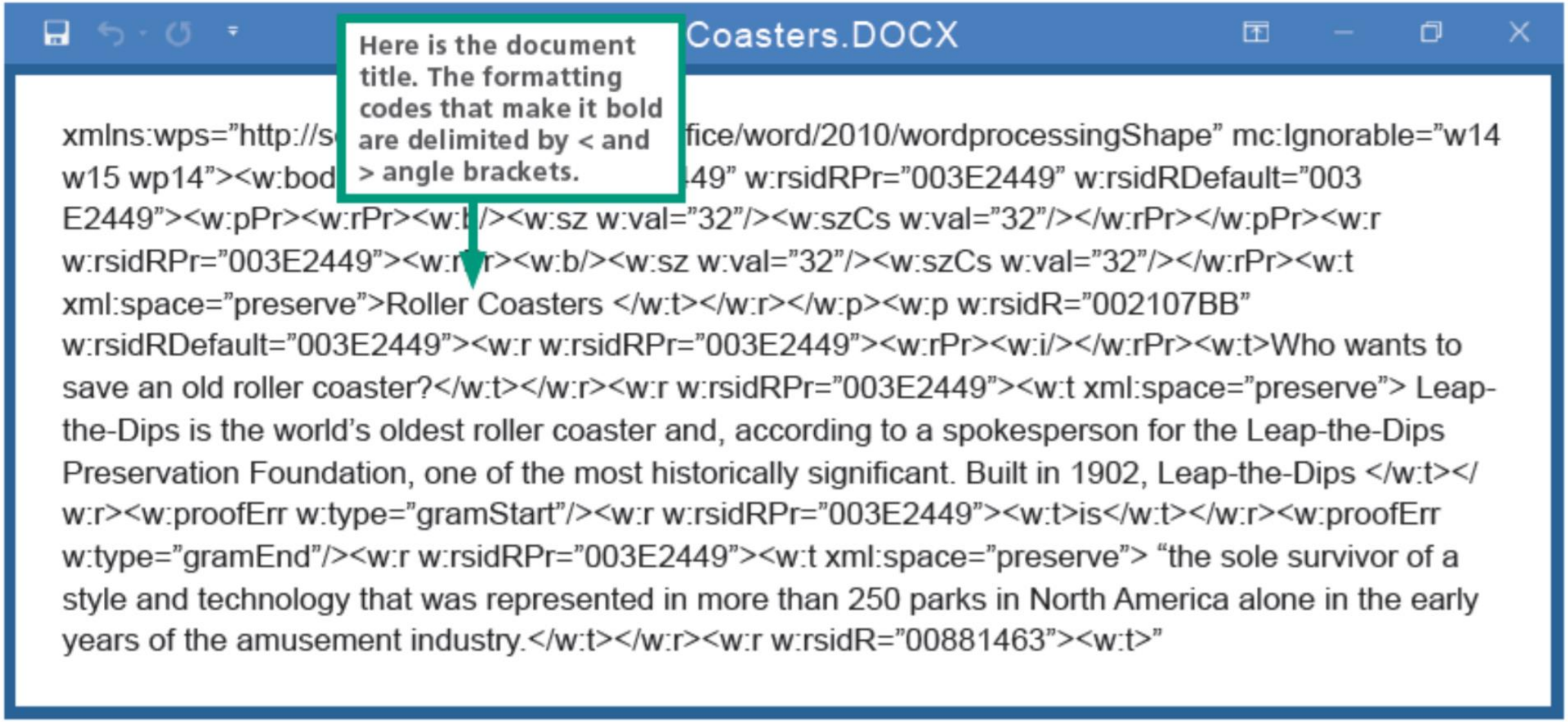
Sample program: https://onlinegdb.com/zSG4LNMu4

ASCII Applications

- Used for numerals
  - Social Security numbers
  - Phone numbers
- Plain, unformatted text (ASCII text) stored in text file (.txt)
  - Apple: “Plain Text”
  - Windows: “Text Document”

Formatted Text Files

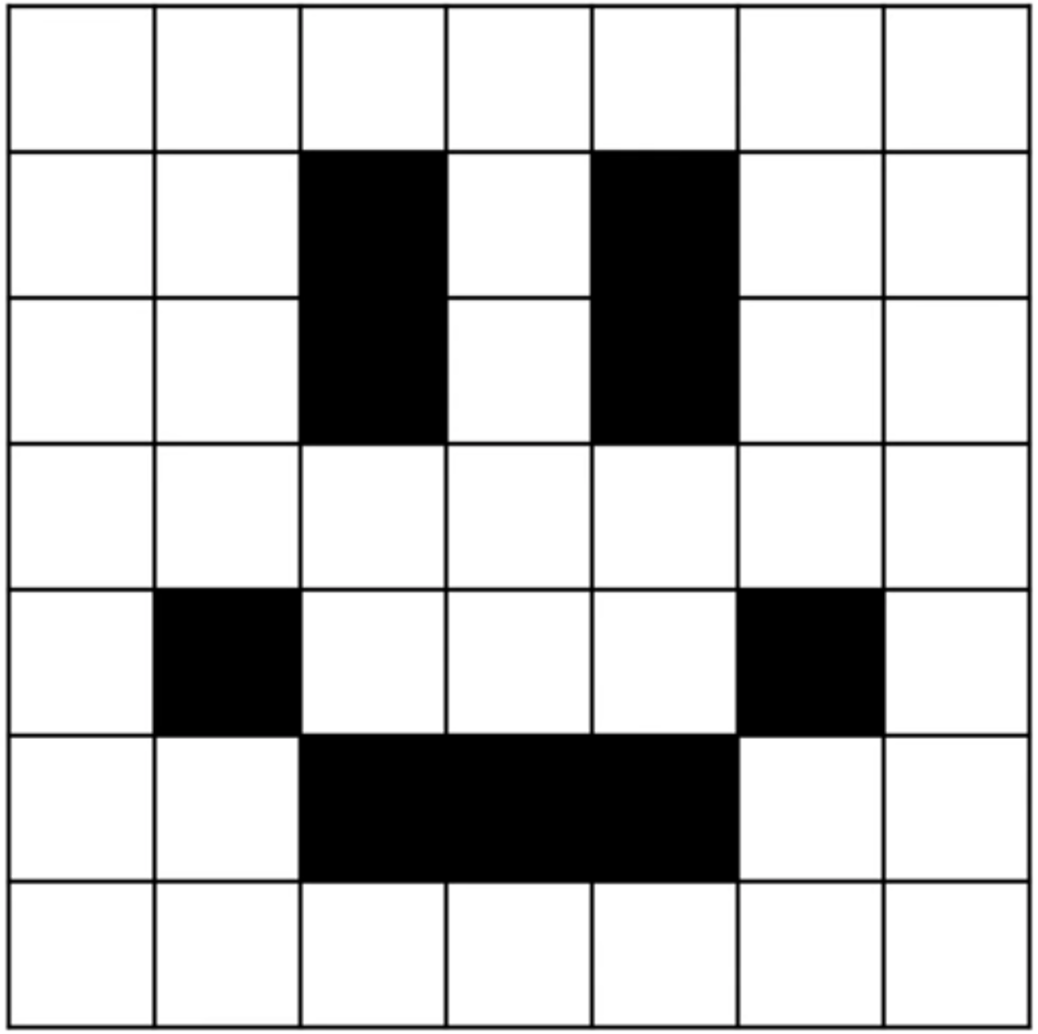
- To create documents with styles and formats, formatting codes must be embedded in the text.
  - Microsoft Word (DOCX)
  - Apple Pages (PAGES)
  - Adobe Acrobat (PDF)
  - Etc.


Coasters.DOCX
Here is the document title. The formatting codes that make it bold are delimited by < and > angle brackets.
xmlns:wps="http://s fice/word/2010/wordprocessingShape" mc:Ignorable="w14 w15 wp14"><w:bod 49" w:rsidRPr="003E2449" w:rsidRDefault="003 E2449"><w:pPr><w:rPr><w:b/><w:sz w:val="32"/><w:szCs w:val="32"/></w:rPr></w:pPr><w:r w:rsidRPr="003E2449"><w:r><w:b/><w:sz w:val="32"/><w:szCs w:val="32"/></w:rPr><w:t xml:space="preserve">Roller Coasters </w:t></w:r></w:p><w:p w:rsidR="002107BB" w:rsidRDefault="003E2449"><w:r w:rsidRPr="003E2449"><w:rPr><w:i/></w:rPr><w:t>Who wants to save an old roller coaster?</w:t></w:r><w:r w:rsidRPr="003E2449"><w:t xml:space="preserve"> Leap-the-Dips is the world's oldest roller coaster and, according to a spokesperson for the Leap-the-Dips Preservation Foundation, one of the most historically significant. Built in 1902, Leap-the-Dips </w:t></w:r><w:proofErr w:type="gramStart"/><w:r w:rsidRPr="003E2449"><w:t>is</w:t></w:r><w:proofErr w:type="gramEnd"/><w:r w:rsidRPr="003E2449"><w:t xml:space="preserve"> "the sole survivor of a style and technology that was represented in more than 250 parks in North America alone in the early years of the amusement industry.</w:t></w:r><w:r w:rsidR="00881463"><w:t>"

Representing Images and Graphics

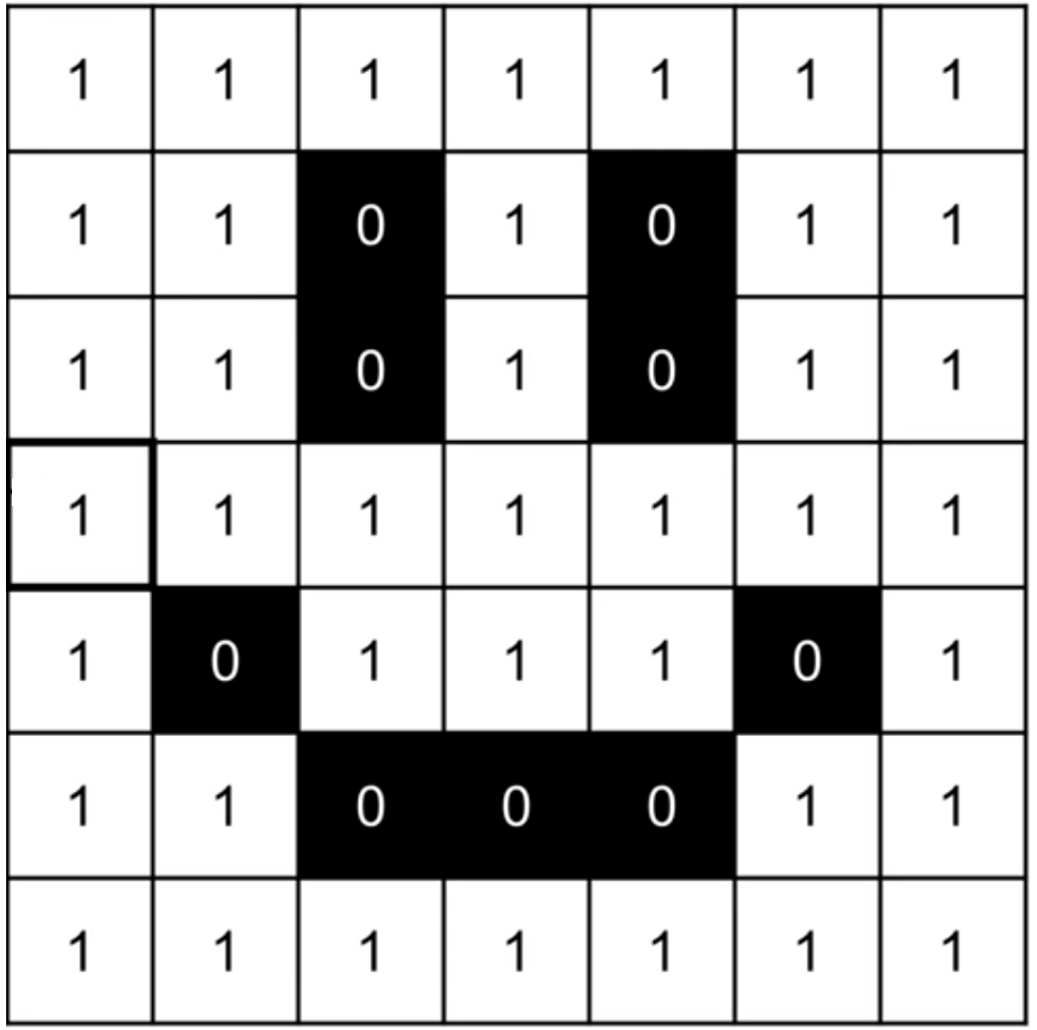
Images

- How could we use binary to represent this image?
- How many colors we need?

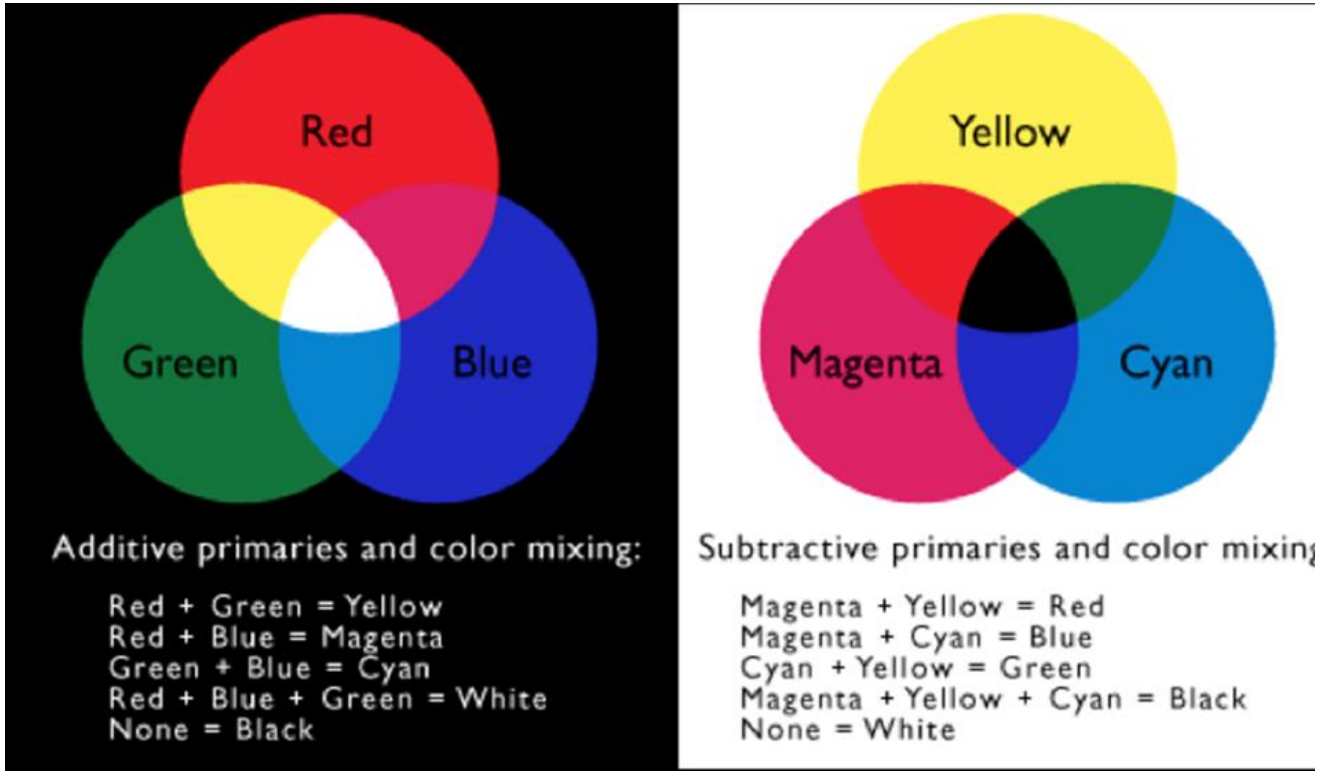


Images

- 2 colors: 1 for white, 0 for black
- A **pixel** (short for "picture element") is the smallest unit of a digital image or display that can be individually manipulated.
- Pixels are the building blocks of any digital image, and each pixel represents a single point of color in the image.



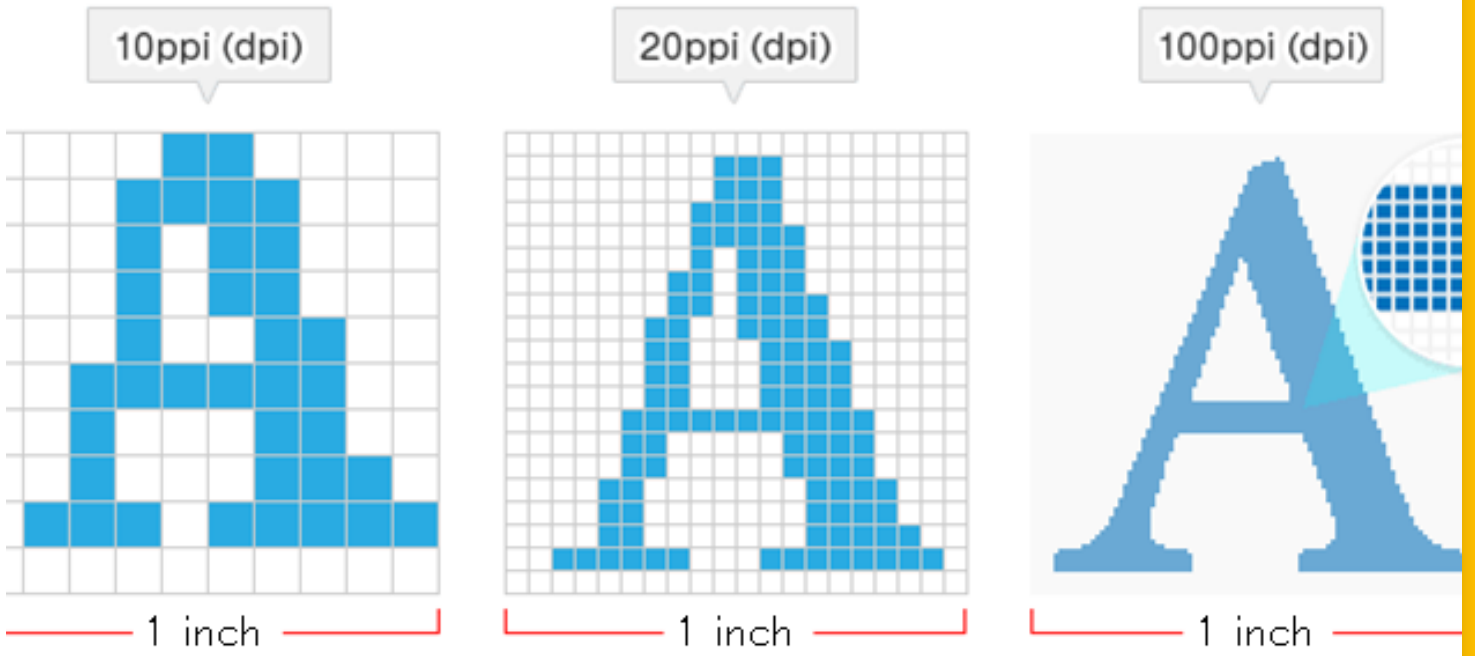
Color



- Color is expressed as a combination of three colors of light
- Each color value is represented as one byte, so each color value ranges from 0 to 255
- Two Types:
  - CMY (cyan-magenta-yellow) → Printers
  - RGB (red-green-blue) → Screen

Resolution

- The resolution of an image refers to the number of pixels it contains.
- Typically expressed as width × height (e.g., 1920 × 1080).
- More pixels mean higher resolution and more detail.

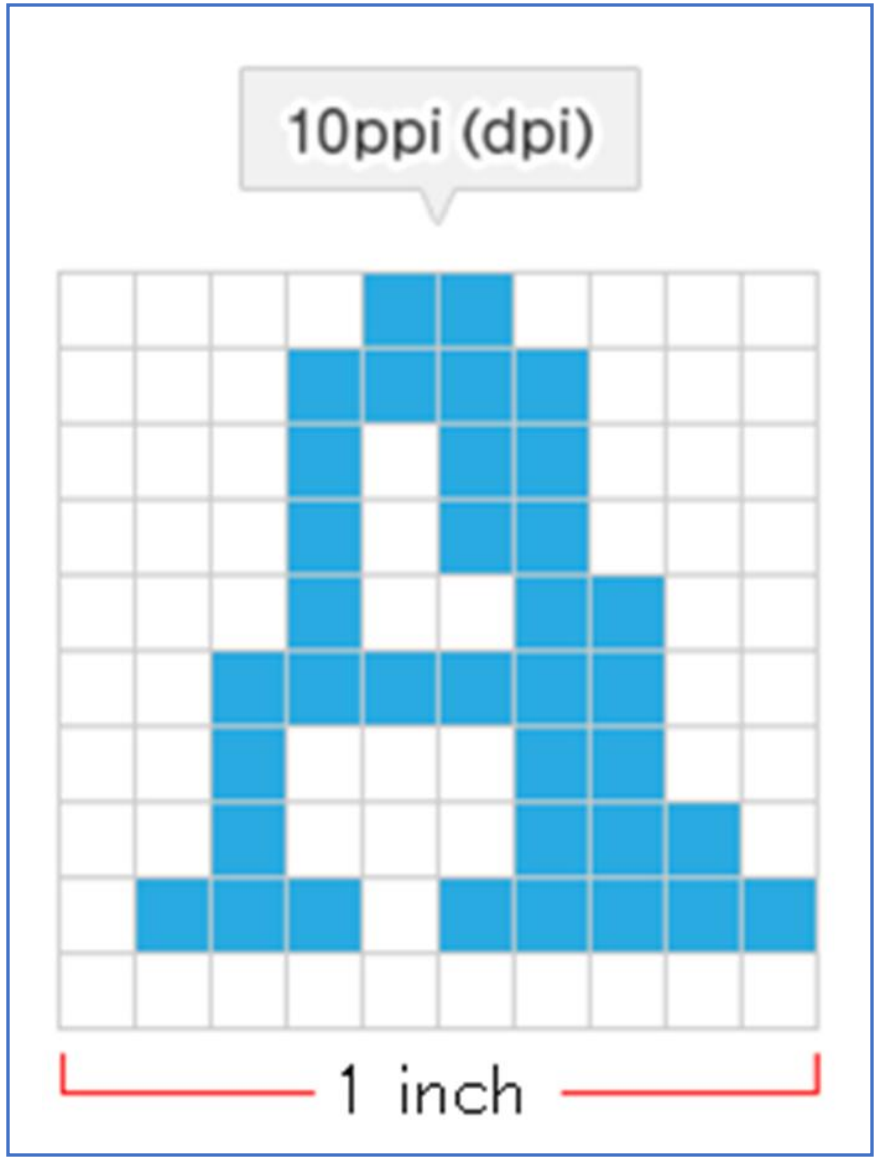


Storage

- Image Specifications:
  - Resolution: 10ppi (10p x 10p)
  - RGB: 3 bytes

How many bytes would be needed to store this image?

  - 10p x 10p = 100 pixels
  - Each pixel has 3 bytes → 100 * 3 = 300 bytes



Medical Imaging

- Magnetic Resonance Imaging (MRI): non-invasive imaging technique that uses strong magnetic fields and radio waves to generate detailed images of soft tissues, organs, and other internal structures.
- Positron Emission Tomography (PET): allows for the observation of metabolic processes in the body.
- Computed Tomography (CT scans): use X-rays to create cross-sectional images of the body.
- Ultrasound: Uses high-frequency sound waves to create real-time images of the body's internal structures.
- X-rays: used to view bones, detect fractures, and identify lung conditions.

Medical Imaging

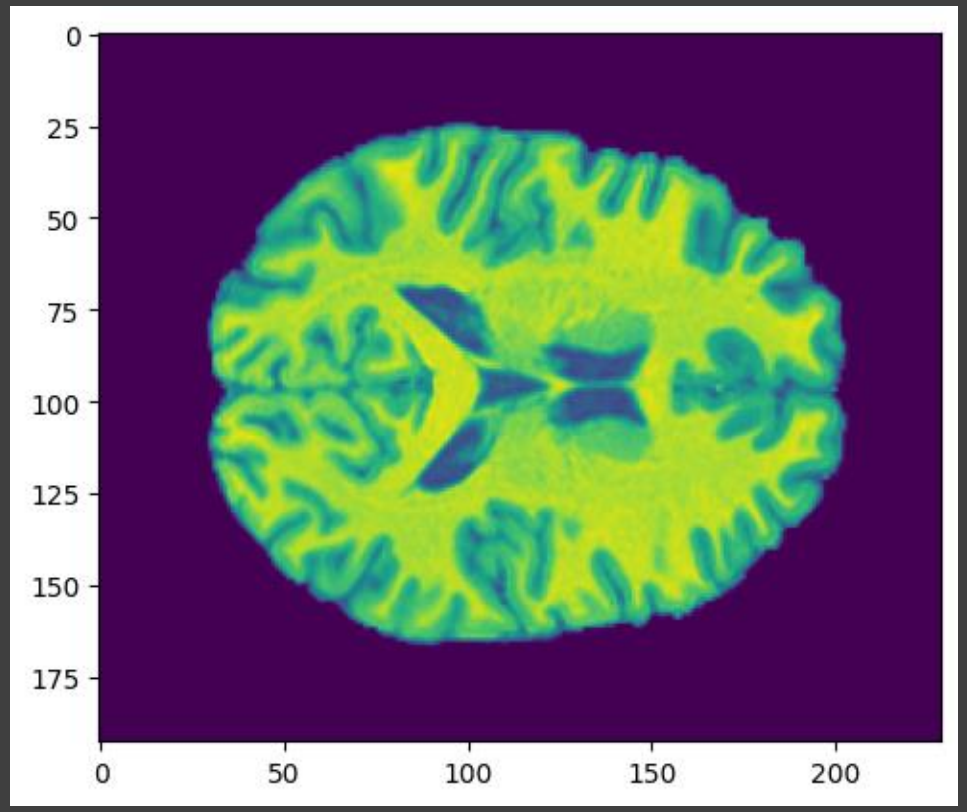
- Each pixel or voxel in the image corresponds to a specific part of the anatomy, with colors and intensities reflecting different tissues and structures.
- A **voxel** (short for "volume element") is the three-dimensional equivalent of a pixel.
- While a **pixel** represents a single point in a 2D image, a voxel represents a point in 3D space

CT Scan

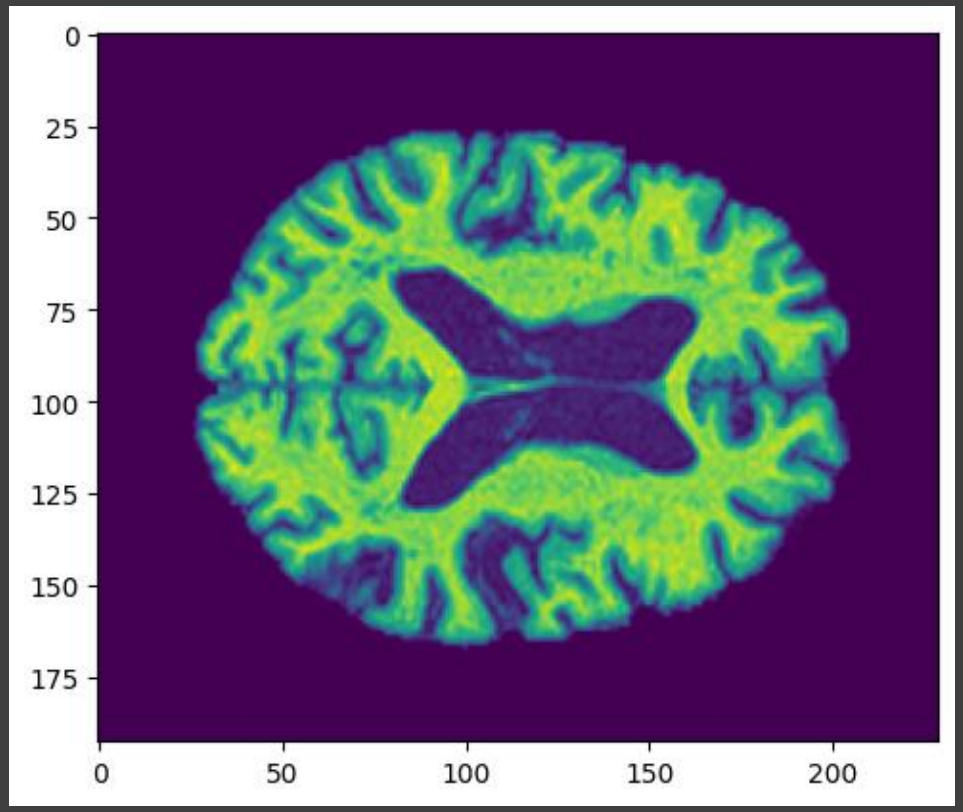
MRI Scan

X-Rays

Ultrasound

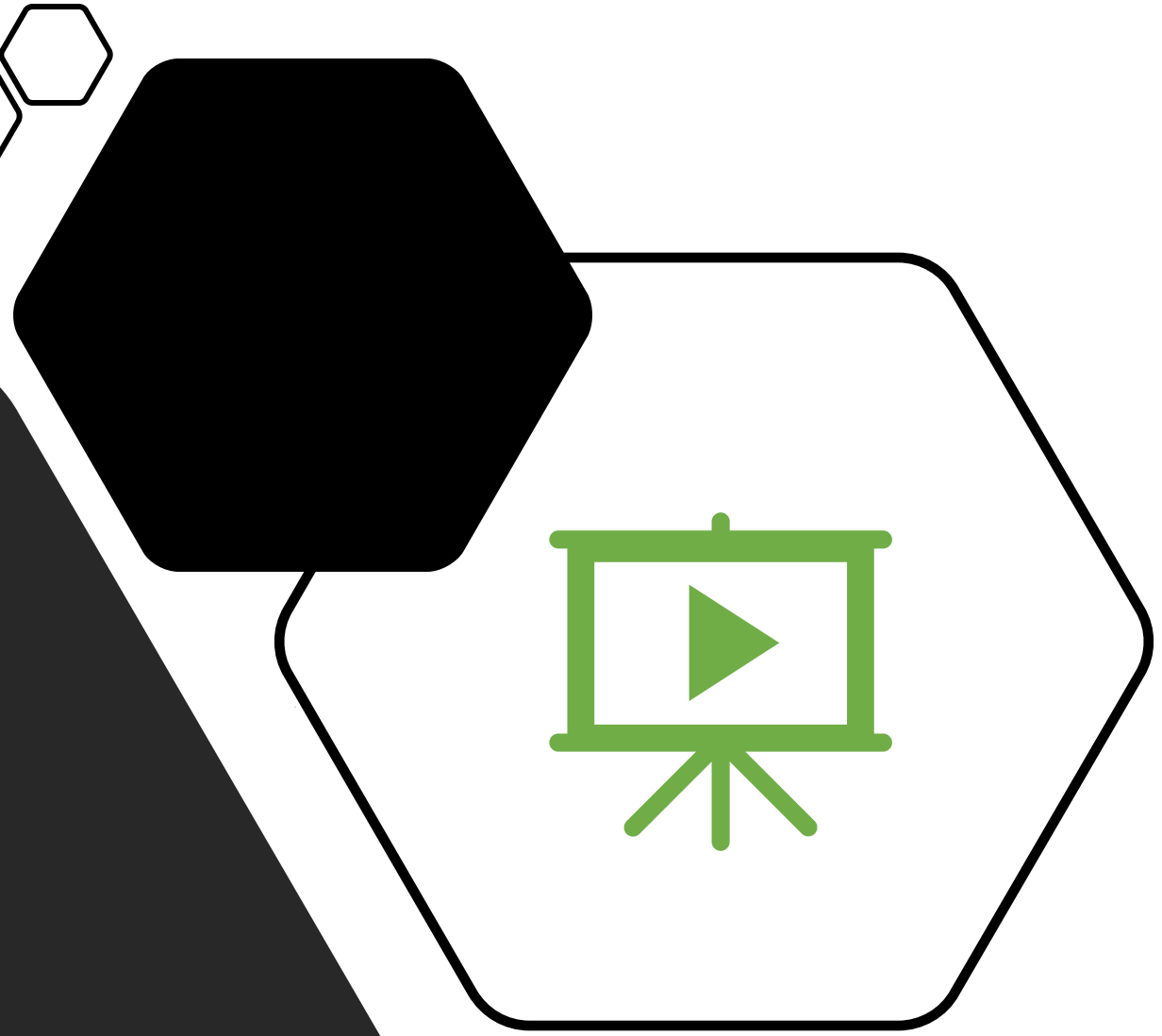


Normal Aging



Alzheimer’s Disease

Representing Video

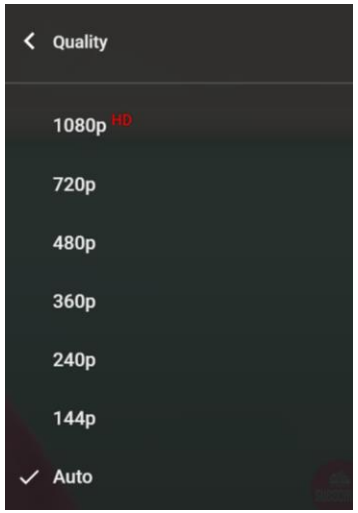


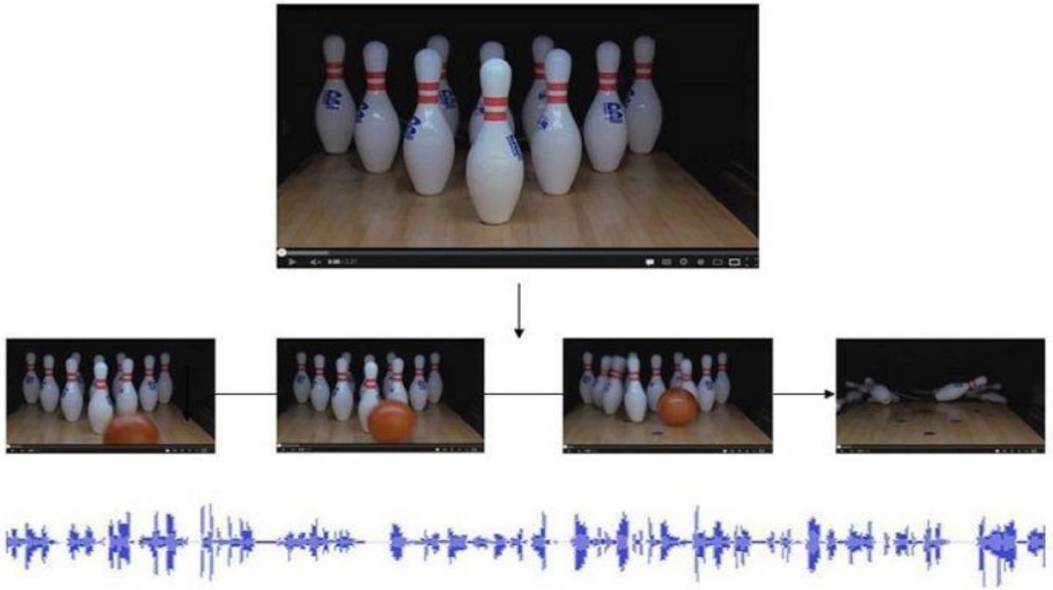
What Is A Video?

- Understood as a series of pictures being “flipped” through with audio
- The same/similar concepts for image and audio representation apply to video

Digital Video

- Made up of individual frames (images), each one representing a time slice of the scene with audio stored as a separate stream
- Frame Rate: frequency at which frames are displayed
- Video Quality:
  - The higher the frame rate the smoother the video will appear
  - The higher resolution the higher the quality of the image



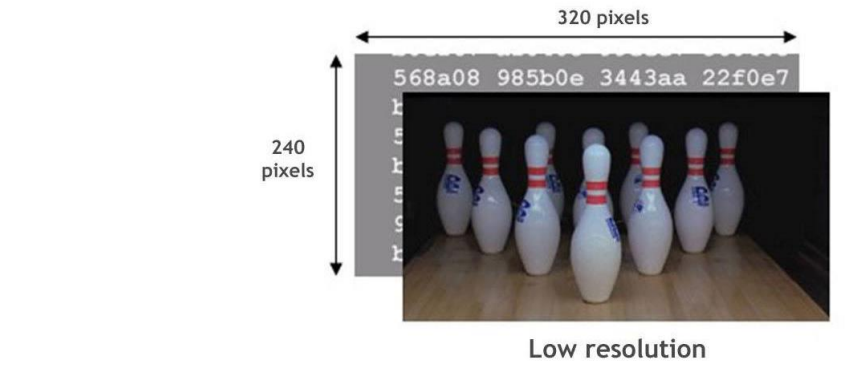


Storage - Example

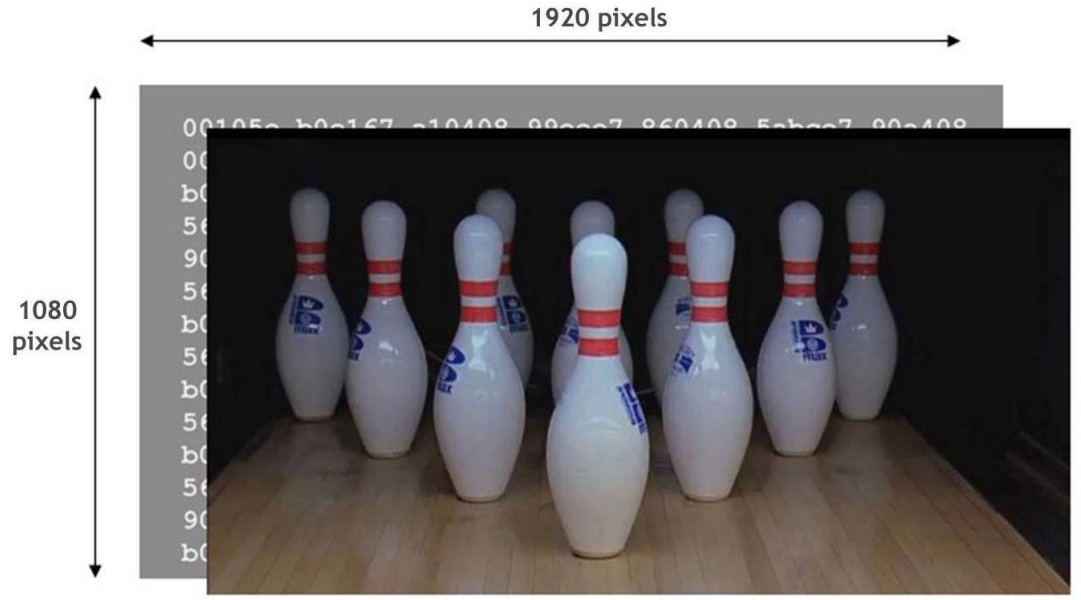
- Video Specifications:
  - Resolution: (Low Resolution) 240p x 320p
  - Frame Rate: 24 frames per second

How many bytes would be needed to store 1 minute of video?

- 240 x 320 = 76,800 pixels in single frame
- Each pixel 3 bytes (RGB) → 230,400 bytes or about 200 KB in a single frame
- 1 second of video → 5,529,600 bytes or 5 MB
- 1 minute of video → 331,776,000 bytes or 332 MB



Low resolution



HD resolution

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